

Li Lai

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CV updated: 2026-6-2

Work Experience

- 2025–Present **Xiamen University**, School of Mathematical Sciences.
Associate Professor
- 2023–2025 **Peking University**, Beijing International Center for Mathematical Research.
Postdoc, Mentor: Liang Xiao
- 2020–2021 **Fudan University**, School of Mathematical Sciences.
Research Assistant, Mentor: Yijun Yao

Education

- 2021–2023 **Tsinghua University**, Department of Mathematical Sciences, Beijing, China.
Ph.D. Mathematics, Advisor: Pin Yu
- 2014–2020 **Tsinghua University**, Department of Mathematical Sciences, Beijing, China.
M.S. Mathematics, Advisor: Pin Yu
- 2010–2014 **Tsinghua University**, Department of Mathematical Sciences, Beijing, China.
B.S. Mathematics

Research Interests

I mainly work in the area of transcendental number theory and Diophantine approximation. I am particularly interested in odd zeta values $\zeta(2k+1)$, p -adic odd zeta values $\zeta_p(2k+1)$, and multiple zeta values $\zeta(k_1, k_2, \dots, k_r)$.

Selected Publications and Preprints

4. Li Lai and Johannes Sprang,
On the $P(t)$ -adic Littlewood conjecture in odd characteristics,
arXiv:2606.00633
3. Li Lai, Johannes Sprang and Wadim Zudilin,
A note on the irrationality of $\zeta_2(5)$,
arXiv:2505.05005v2
2. Li Lai, Jiong-Yue Li and Pin Yu,
On the rigidity of stationary charged black holes: small perturbations of the non-extremal Kerr-Newman family,
J. Differential Geom. 125 (2023), no. 3, 553–612.
arXiv:1911.10560
1. Li Lai and Pin Yu,
A note on the number of irrational odd zeta values,
Compos. Math. 156 (2020), no. 8, 1699–1717.
arXiv:1911.08458

Awards and Honors

- 2012 Nianzeng Sun Mathematical Analysis Award (at Tsinghua University)
- 2010 51st International Mathematical Olympiad: Gold Medal

Teaching

- Spring 2026 Exercise Course of Mathematical Analysis, Xiamen University
- Fall 2024 Rational Functions and the Irrationality of Odd Zeta Values, Mini Course, Peking University
- Spring 2024 Advanced Mathematics B (2), Peking University
- Spring 2021 Rational Functions and the Irrationality of Odd Zeta Values, Short Course, Fudan University

Seminar/Conference (Co)Organized

- July 27–31, 2026 The 5th International Conference on Multiple Zeta Values and Related Fields, Changsha, China
- Fall 2021–Spring 2022 Tsinghua-BIMSA Learning Seminar on Multiple Zeta Values, Tsinghua University

Other Experiences and Activities

- 2017 Finisher of Columbia168 ULTRA-TRAIL[®] THREE GORGES (168 km trail running)
- Spring 2013 Exchange student at École Normale Supérieure, Paris, France

Publications and Preprints

14. L. Lai and J. Sprang, *On the $P(t)$ -adic Littlewood conjecture in odd characteristics*, arXiv:2606.00633
13. L. Lai, C. Lupu and J. Sprang, *On the irrationality of certain p -adic zeta values*, Res. Math. Sci. **12** (2025), no. 4, Paper No. 77. arXiv:2505.23088
12. L. Lai and J. Li, *A partial result towards the Chowla–Milnor conjecture*, arXiv:2505.12687v2
11. L. Lai, J. Sprang and W. Zudilin, *A note on the irrationality of $\zeta_2(5)$* , arXiv:2505.05005v2
10. L. Lai, *A note on the number of irrational odd zeta values, II*, to appear in J. Théor. Nombres Bordeaux, arXiv:2501.05321v2
9. L. Lai, *Small improvements on the Ball–Rivoal theorem and its p -adic variant*, arXiv:2407.14236v2
8. L. Lai and J. Sprang, *Many p -adic odd zeta values are irrational*, to appear in Michigan Math. J., arXiv:2306.10393v2
7. L. Lai, *On the irrationality of certain 2-adic zeta values*, Int. J. Number Theory, **21** (2025), no. 1, 207–235. arXiv:2304.00816
6. S. Charlton, H. Gangl, L. Lai, C. Xu and J. Zhao, *On two conjectures of Sun concerning Apéry-like series*, Forum Math., **35** (2023), no. 6, 1533–1547. arXiv:2210.14704
5. L. Lai, C. Lupu and D. Orr, *Elementary proofs of Zagier’s formula for multiple zeta values and its odd variant*, Proc. Amer. Math. Soc., **154** (2026), no. 01, 11–24. arXiv:2201.09262
4. L. Lai, *On the largest prime divisor of $n! + 1$* , to appear in Bull. Aust. Math. Soc., arXiv:2103.14894
3. L. Lai and L. Zhou, *At least two of $\zeta(5), \zeta(7), \dots, \zeta(35)$ are irrational*, Publ. Math. Debrecen, **101**/3–4 (2022), 353–372. arXiv:2103.00904
2. L. Lai, J.-Y. Li and P. Yu, *On the rigidity of stationary charged black holes: small perturbations of the non-extremal Kerr–Newman family*, J. Differential Geom. **125** (2023), no. 3, 553–612. arXiv:1911.10560
1. L. Lai and P. Yu, *A note on the number of irrational odd zeta values*, Compos. Math., **156** (2020), no. 8, 1699–1717. arXiv:1911.08458